

EEE4114F: Problem Set 3

(z-transform)

1. The z-transform of the sequence $x[n]$ is

$$X(z) = \frac{z + 2z^{-2} + z^{-3}}{1 - 3z^{-4} + z^{-5}}.$$

If the ROC includes the unit circle, find the DTFT of $x[n]$ at $\omega = \pi$.

2. Find the z-transform of each of the following sequences:

(a) $x[n] = 3\delta[n] + \delta[n - 2] + \delta[n + 2]$.

(b) $u[n] - u[n - 10]$.

3. Find the z-transform of each of the following sequences. Wherever convenient, use the properties of the z-transform to make the solution easier:

(a) $x[n] = (1/3)^n u[-n]$.

(b) $(1/2)^n u[n + 2] + 3^n u[-n - 1]$.

(c) $(1/3)^n \cos(n\omega_0) u[n]$.

(d) $\alpha^{|n|}$.

4. Use the derivative property to find the z-transform of the following sequences:

(a) $x[n] = n(\frac{1}{2})^n u[n - 2]$

(b) $x[n] = \frac{1}{n}(-2)^{-n} u[-n - 1]$.

5. Evaluate the convolution of the two sequences $h[n] = (0.5)^n u[n]$ and $x[n] = 3^n u[-n]$.

6. Using the derivative property, find the z-transform of $x[n] = |n|(1/2)^{|n|}$.

7. Let $x[n]$ be a left-sided sequence that is zero for $n > 0$. If

$$X(z) = \frac{3z^{-1} + 2z^{-2}}{3 - z^{-1} + z^{-2}},$$

find $x[0]$.

8. Find the inverse z-transform of the following:

(a) $X(z) = 4 + 3(z^2 + z^{-2})$, $0 < |z| < \infty$.

(b) $X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{3}{1 - \frac{1}{3}z^{-1}}$, $|z| > \frac{1}{2}$.

(c) $\frac{1}{1 + 3z^{-1} + 2z^{-2}}$, $|z| > 2$.

(d) $\frac{1}{(1 - z^{-1})(1 - z^{-2})}$, $|z| > 1$.

9. Find the inverse z-transform of

$$X(z) = \frac{1 + \frac{1}{4}z^{-1}}{(1 - \frac{1}{2}z^{-1})^2}, \quad |z| > \frac{1}{2}.$$

10. Use the z-transform to perform the convolution of the following two sequences:

$$x[n] = \delta[n] - 2\delta[n - 2]$$

$$h[n] = 2\delta[n] - 2\delta[n - 1] + 3\delta[n - 2] + \delta[n - 3].$$